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Learning Personalized Treatment Strategies with Predictive and Prognostic Covariates in Adaptive Clinical Trials

Andres Alban,[a,*] Stephen E. Chick,[b] Spyros I. Zoumpoulis[c]

[a] Frankfurt School of Finance & Management, 60322 Frankfurt, Germany; [b] Technology and Operations Management, INSEAD, 77300 Fontainebleau, France; [c] Decision Sciences, INSEAD, 77300 Fontainebleau, France

*Corresponding author

Contact: a.alban@fs.de, https://orcid.org/0000-0002-9552-4039 (AA); stephen.chick@insead.edu, https://orcid.org/0000-0002-8026-1571 (SEC); spyros.zoumpoulis@insead.edu, https://orcid.org/0000-0002-4662-7787 (SIZ)

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Abstract. Biomedical research may uncover insights regarding the interaction of the treatments of a disease with patient covariates. We show how to use such insights to improve the efficiency of adaptive clinical trials for precision medicine by extending ideas from optimal Bayesian learning. We present a model for response-adaptive multi-arm clinical trials that leverages knowledge about the predictive-prognostic covariate structure to accelerate the learning of personalized treatment strategies that obtain the best expected outcomes for post-trial patients. Our base model is a contextual linear bandit for best-arm identification, and outcomes may be observed with delay. We characterize the optimal policy for sequentially allocating treatments to in-trial patients and, because it is hard to compute, propose several computable heuristics based on Bayesian one-step look-ahead techniques. We prove that several of our proposed heuristics are asymptotically optimal in learning treatment strategies. Numerical results based on two case studies motivated by sepsis management show that our heuristics can significantly improve clinical trial efficiency to learn a treatment strategy for precision medicine. We provide extensions that allow for rewards from outcomes of in-trial patients (resolving the exploration-exploitation tradeoff) and for inferring covariate structure using Lasso when biomedical insights on covariate structure are lacking. Our proposed trial design is of interest to funders, designers, and managers of clinical trials. It may also apply to other contextual bandit problems in settings where insights about covariate-treatment interactions are available.

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Keywords: adaptive clinical trial design • contextual bandit • sequential learning • Bayesian optimization • expected value of information • sepsis

1. Introduction

Precision medicine tailors treatments to patient characteristics and has received considerable attention in recent years (NIH 2022, U.S. FDA 2023). It has provided significant health improvements in cancer treatment (Tsimberidou et al. 2020) and is expected to lead to advancements in other areas, such as cardiovascular disease (Tada et al. 2021), neurology (Cutter and Liu 2012), and sepsis (Rello et al. 2018, Seymour et al. 2019). One key aspect of developing personalized treatments is the design of clinical trials to find the best intervention for each type of patient efficiently (Berry 2011, Opal et al. 2014, Schork 2018).

This paper explores how adaptive clinical trials can leverage biomedical insights about the interaction of patient covariates (characteristics) with treatments to learn the best *treatment strategy* faster. A treatment strategy assigns a treatment based on the patient's covariates, and "best" is defined as maximizing the mean outcome. Specifically, we focus on whether biomedical insights on the *predictive* and *prognostic* nature of covariates can improve trial efficiency. Predictive covariates are those that interact with treatments and are key to identifying groups of patients that benefit from personalized treatment. Prognostic covariates affect the outcome without treatment interactions and help improve the estimates of the benefit of personalized treatment (Oldenhuis et al. 2008, U.S. FDA 2023). Predictive covariates are sometimes called moderators (Wu and Zumbo 2008) and prognostic covariates have been called action-independent effects (Carranza et al. 2023).

Biomedical insights regarding covariate-treatment interactions may be available for at least two reasons. One reason is that observational data may suggest

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that one or more potential interactions are worth exploring. For example, the MARS Consortium (Scicluna et al. 2017) analyzed blood genomic data from sepsis patients and found that they can be clustered into four groups (endotypes). A follow-up analysis (van Mourik et al. 2022) suggested the potential presence of a predictive interaction—patients in one cluster seemed to respond better to certain sepsis treatment plans than others—alongside known prognostic covariates. A second reason is that biomedical research may focus on designing new therapies to target patient subgroups identified by biomarkers (e.g., in oncology; Tsimberidou et al. 2020). Such potential predictive interactions, based on observational data or biomedical insights, must then be validated, often by clinical trials.

This motivates our *key question*, which is relevant for funders, designers, or managers of clinical trials: Can optimal learning techniques improve the efficiency of finding the best treatment strategy in clinical trials in settings where insights about the predictive-prognostic covariate structure are available? We ask this for multi-arm adaptive clinical trials that allocate new trial participants to treatment arms based on their covariates and on the information accumulated during the trial so far. We answer affirmatively.

Other applications, beyond clinical trials, include (i) online advertising, where the alternative ads target specific characteristics of the users, whereas some (prognostic) user characteristics may affect the outcome (e.g., click or not) independent of the ad; and (ii) engineering testing, where the realization of the environment represents the arrival of subjects and the alternative system configurations represent the treatments.

1.1. Overview of Model

We develop a mathematical model of a Phase 3 clinical trial for precision medicine augmented by insights about prognostic and predictive covariates (Section 2). The trial manager sequentially enrolls a fixed number of patients and selects a treatment from the trial arms to assign to each patient after observing the patient's covariates. Based on information collected in the trial so far, the trial manager makes treatment assignments with the goal of efficiently identifying a treatment strategy that maximizes the reward for a patient population after the trial is completed. We assume patient outcomes are normally distributed. The unknown mean is determined by a linear response model with parameters that depend on whether covariates are suspected to be predictive, prognostic, or both. The trial manager uses patient outcomes to update sequentially the posterior distribution of the parameters of the linear response model with Bayes' rule.

The base model is a contextual multi-armed bandit, where the context is the vector of patient covariates, with *offline rewards*: Outcomes from patients during the trial inform the selection of a treatment strategy that is then used to treat a post-trial patient population. Our model also includes two features of clinical trials that may differ from other contextual learning settings: Outcomes may be observed with a delay, and the covariates of the next trial participant are random rather than chosen by an analyst.

We give two important extensions to the base model. First, Section 6 accounts for *online* or *combined online and offline rewards*, to also account for the outcomes of patients in the trial. Second, we observe how inference can degrade if the likely predictive-prognostic covariate structure is not known and provide a mechanism that uses Lasso (Tibshirani 1996) to infer a labeling of covariates as predictive or prognostic for our model (Section 9).

1.2. Summary of Contributions

1.2.1. Modeling. We present what appears to be the first fully sequential learning model that accounts for both predictive and prognostic covariates with Bayesian techniques, as well as offline and online rewards. Other sequential learning work often assumes that covariates are all predictive (involving more parameters to estimate), all prognostic (missing the chance to learn which treatment is best for which subpopulation), or not relevant. We model multi-arm trials with correlated mean rewards across arms (similar treatments may have similar effects) and patient types (similar patients may have similar responses to a treatment) and delayed outcomes.

1.2.2. Algorithms. Although the optimal policy exists and can be characterized by Bellman's equation for a dynamic program (Section 3), its computation suffers from the curse of dimensionality. We present four heuristics for assigning patients to treatment arms during the trial (Section 4). To the best of our knowledge, this is the first work to adapt the Bayesian one-step look-ahead approach for sequential learning (e.g., the expected value of information (EVI) of Chick and Inoue (2001) or knowledge gradient (KG) of Frazier et al. (2008)) to give arm-allocation indices in the presence of predictive and prognostic covariates and a delay in observing outcomes after treatment decisions. We call the main proposed heuristic fEVI because it maximizes the EVI of myopic look-ahead allocations of patients to treatments to learn the best treatment strategy, a function (f) from covariates to treatments. The indices may be hard to compute when delays are positive or when covariates are continuous-valued. We offer a heuristic that computes these indices with Monte Carlo (MC) simulation and a generally useful variance reduction that builds on Frazier et al. (2009). We also provide heuristics that allow a trial manager to control the degree of randomization in a trial.

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1.2.3. Analysis. We show that all but one of our heuristics are asymptotically optimal (Section 5). We show that the remaining heuristic is asymptotically optimal in some important special cases. The justification of the theoretical guarantees involves some novel variations on standard EVI/KG proof structures (Online Appendix EC.3).

1.2.4. Pragmatic Contributions. We adapt ideas of Powell and Ryzhov (2012) to help specify the requisite prior distributions for unknown regression parameters of predictive and prognostic covariates (Section 7). We then assess the proposed heuristics on simulated clinical trials motivated by sepsis management, against comparator heuristics from the contextual bandit, contextual ranking and selection (R&S), or biostatistical streams (Sections 8 and 9). Some comparators are exploration focused, and others balance exploration with exploitation. Numerical results validate that our approach has promise in clinical trial designs for precision medicine when insights about the predictive-prognostic covariate structure are available. Our results also validate that the proposed MC-based indices are useful and that our heuristic for inferring covariate structure, when it is unknown, can support faster learning relative to assuming all treatments can potentially interact with all covariates.

We delineate some areas that lie outside of our scope in this work. We propose promising new techniques for clinical trial design, but running such a trial is beyond our scope here. We assume that outcomes (or rewards) for patients are real-valued. Bernoulli trials are not formally analyzed. We do not handle multiple longitudinal outcome measures through time (as do Anderer et al. (2022)). The delays considered here are short enough relative to the enrollment period to allow adaptations in treatment allocation. Multiyear survival studies are less suited to the model as proposed. We do not consider covariates of interest that are observable with a delay or that evolve through time. We discuss adaptations of the model to handle further practical issues and areas of future research in the Online Appendix.

1.3. Related Literature

This work is linked to several streams of literature: work on using covariates in clinical trial design and data analysis, work on sequential sampling in the contextual bandit and R&S streams, and work on structural assumptions about the functional form of the mean response as a function of covariates.

1.3.1. Covariates and Clinical Trials. The use of clinical trial data for identifying subgroups of patients that have an exceptional response to treatment has been studied under the subgroup selection area of biostatistics (Hu and Rosenberger 2006, Foster et al. 2011, Lipkovich et al. 2017). Interesting approaches include information theory (Sechidis et al. 2018), which ranks covariates by their predictive and prognostic value, and optimization (Bertsimas et al. 2019b). That literature focuses on analyzing already collected clinical trial data to identify subgroups, whereas we aim at designing trials that choose how to allocate patients to treatments to more efficiently learn treatment strategies.

The use of prognostic covariates for clinical trials focuses on balancing covariates across the treatment arms, most notably using the "biased coin" approach (Pocock and Simon 1975), which adjusts the randomization probabilities to favor the balancing of covariates. Ma et al. (2024) provide a more recent and general framework for two-armed trials. Such designs can adapt the randomization based on patient responses (Zhang et al. 2007, Hu et al. 2015). Optimization approaches have been proposed (Bertsimas et al. 2019a, Bhat et al. 2020). That line of research assumes that all covariates are prognostic, aiming to learn average treatment effects in a population, and does not tailor treatment to patients.

The literature on trial design for precision medicine is expanding (Berry 2011, Lai et al. 2013, Wang and Yee 2019). Initial work focused on predictive covariates, but increasing attention is given to both predictive and prognostic effects (Symmans et al. 2018, U.S. FDA 2023), our focus here. Zhao et al. (2022) propose a framework to balance prognostic covariates while using a response-adaptive proportion of allocations to each arm based on predictive covariates: We adapt that work here as a comparator. We distinguish the learning of personalized treatments in our small data, clinical trial context from work with big, pooled data and/or Markov decision processes to personalize dynamic interventions through time (Tunc et al. 2014, Ayer and Chen 2018, Chen et al. 2022).

1.3.2. Sampling to Learn the Best Alternative. The R&S literature (Kim and Nelson 2006, Branke et al. 2007) focuses on finding the best alternative among a small set focusing on offline rewards. This rich literature primarily focuses on selecting the best overall treatment rather than a treatment strategy. More recently, R&S has been extended to account for covariates (Pearce and Branke 2018, Gao et al. 2019, Xiong 2020, Ding et al. 2021, Shen et al. 2021, Li et al. 2022). These papers assume control over both covariates and treatments for various simulation optimization problems. In contrast, our work assumes that the covariates are random arrivals, as is typical in clinical trials.

Another related stream of literature studies the multi-armed bandit problem (Auer 2002). The contextual bandit literature has developed policies to minimize cumulative regret asymptotically (Rigollet and

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Zeevi 2010, Goldenshluger and Zeevi 2013, Russo and Van Roy 2014, Villar and Rosenberger 2018, Bastani and Bayati 2020, Bastani et al. 2021). Other contextual multi-armed bandit work studies the best-arm identification problem to maximize offline rewards (Qin and Russo 2024). Russo (2020) proposes variations of Thompson sampling that are more amenable for offline rewards. Although not explicitly modeled in previous work, Thompson sampling and its variations can be used with our Bayesian framework to account for predictive and prognostic covariates, and we use them as comparators. Our focus is mainly on offline rewards obtained upon committing to a treatment strategy for future patients, but we also show how to account for online rewards in our framework. We will use "expected opportunity cost" to refer to suboptimality of treatment strategies for offline rewards (post-trial patients) and "cumulative regret" to refer to suboptimality for online rewards (in-trial patients).

The contextual R&S and bandit works above consider all covariates to be predictive and do not exploit the potential gains in learning quicker using prognostic biomarkers. A recent paper (Carranza et al. 2023) decomposes online rewards of a contextual bandit into treatment effect and confounder terms (analogous to our model below of predictive and prognostic covariates) and gives asymptotic frequentist bounds on cumulative regret.

Our approach extends the EVI/KG framework of Bayesian optimization (Chick and Inoue 2001, Frazier et al. 2008, Powell and Ryzhov 2012) to allocate patients to arms in a fully sequential manner before selecting a treatment strategy. Those works do not model covariates. Some contextual R&S work also uses a similar EVI approach (Pearce and Branke 2018, Ding et al. 2021). We model correlated outcomes across treatment types, patient types, or both, building on Frazier et al. (2009) and Chick et al. (2022). We extend Alban et al. (2021) with several model extensions and with theoretical results.

Our model is related to other KG work as well. Ryzhov and Powell (2011) study a problem where the set of implementation decisions (choice of a path in a graph) are not in one-to-one correspondence with the set of sampling choices (measure an arc of the graph). Similarly, our model's implementation decision (choice of a function from covariates to treatment options) differs from the sampling actions (choice of treatment for a patient). We build on Negoescu et al. (2011) for learning regression coefficients by incorporating structural knowledge about the predictive and prognostic nature of covariates. Chick and Inoue (2001) account for batching with stochastic outputs but use a less precise EVI approximation and do not model covariates. Wu and Frazier (2016), Wang et al. (2020), and Astudillo et al. (2021) explore batch and parallel allocations with Bayesian optimization. Those works share a commonality with our delayed observation model: Multiple allocations are made before observing outcomes. We differ in that they can choose covariate values for the batch simultaneously, whereas our covariates arrive randomly and sequentially, and our observations are stochastic rather than deterministic. Finally, Wang et al. (2016) studied prognostic (but not predictive) covariates for 0–1 (rather than real-valued) outcomes.

Other related works on clinical trial design that do not focus specifically on covariates but that relate to our sequential learning include Williamson et al. (2017), Jacko (2018), Pallmann et al. (2018), and Rojas-Cordova and Bish (2018).

To handle delays, some work has shown that ignoring allocations to subjects whose outcomes are yet to be observed when making further allocations does not significantly affect asymptotic properties (e.g., Joulani et al. (2013) in the bandit literature; Bai et al. (2002) and Hu and Zhang (2004) for urn models in clinical trials). We empirically show that accounting for such allocations can be valuable in the short term (Section 8.4).

1.3.3. Modeling the Response. There are several approaches to modeling the *response function*, the expected outcome of a patient with a given set of covariates and treatment. If the space of covariates and treatments is finite, one might allow each response for each covariate-treatment combination to be learned independently and effectively model free (Lai et al. 2013, Villar and Rosenberger 2018, Gao et al. 2019, Li et al. 2022). Those works are a special case of our model: They assume that the covariates are predictive and discrete.

We assume a linear response function, which is a common model in the literature and tractable (Goldenshluger and Zeevi 2013, Bhat et al. 2020, Ding et al. 2021, Shen et al. 2021). We use Bayesian linear regression to learn the coefficients of the model (Negoescu et al. 2011). Instead of learning a different model for each treatment (a standard assumption in multi-armed bandits; Bastani et al. 2021), we learn a single model in which covariates can be predictive and prognostic and arms may have correlated mean rewards.

2. Mathematical Model of a Sequential Trial with Covariates

We formulate a model of an adaptive multi-arm clinical trial that enrolls a fixed number of patients sequentially through time. The trial manager assigns a treatment for each enrolled patient after observing the patient's covariates. We model the patient outcomes using a linear response function that allows for treatment effects, effects from predictive and prognostic covariates, and homoscedastic noise. The trial manager uses the patient outcomes, observable with some delay, to update sequentially in a Bayesian manner the posterior distribution for the parameters of the

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linear response model. The trial manager assesses all the information collected thus far to decide on each subsequent patient's treatment allocation. After the trial, the trial manager has learned a treatment strategy, that is, a mapping from patient covariates to treatments, which the manager then applies to a population of post-trial patients. The trial manager's objective is to learn the best treatment for each set of covariates in order to optimize the expectation of outcomes for a population of post-trial patients. Table EC.1 in the Online Appendix summarizes the notation of the model.

2.1. Information and Decisions

Consider a trial with a budget to enroll a fixed number of patients, T. The patients are enrolled sequentially in evenly spaced time steps, where the time steps $t = 0, 1, \ldots, T$ represent the number of patients that have been allocated to treatment. At time steps $t = 1, 2, \ldots, T$, the trial manager observes one patient with covariates $\mathbf{X}_t \in \mathscr{X} \subseteq \mathbb{R}^m$ and allocates treatment $W_t \in \mathscr{W} = \{1, 2, \ldots, n\}$ to that patient.

The patient's outcome $Y_t \in \mathscr{Y} \subseteq \mathbb{R}$ is observed after a fixed delay of $\Delta \in \{0, 1, \ldots, T-1\}$ time steps, so the trial manager observes at time $t + \Delta$ the outcome of the patient who was enrolled at time t. At time steps $t = T+1, T+2, \ldots, T+\Delta$, no more allocations are made but patient outcomes are still observed. The response function $r_{\boldsymbol{\mu}}(\mathbf{X}_t, W_t) = \mathbb{E}[Y_t \mid \boldsymbol{\mu}, \mathbf{X}_t, W_t]$ yields the expected outcome of the patient given an unknown vector of parameters $\boldsymbol{\mu}$, the covariates, and the treatment. Figure 1 summarizes the sequence of events.

At time $t = T + \Delta$ the trial concludes and each of P post-trial patients with covariates $\tilde{\mathbf{X}}_i \in \mathscr{X}$ receives treatment $\tilde{W}_i \in \mathscr{W}$ and obtains health outcome $\tilde{Y}_i$, where $i = 1, 2, \ldots, P$ indexes the post-trial patients. We assume here that P is a known constant. The expected outcome for post-trial patients is also assumed to be $r_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_i, \tilde{W}_i) = \mathbb{E}[\tilde{Y}_i \mid \boldsymbol{\mu}, \tilde{\mathbf{X}}_i, \tilde{W}_i]$. The distribution of the covariates may differ for patients in the trial, $\mathbf{X}_t$, and those post-trial, $\tilde{\mathbf{X}}_i$, as discussed below in Section 2.4.

The *history* $\mathbf{H}_t$ represents all the available data at time t. At time $t = 0$, the trial manager has not observed any patients, so $\mathbf{H}_0 = \emptyset$. For $1 \le t \le \Delta$ no patient outcomes have been observed yet, so $\mathbf{H}_t$ includes only the covariates and treatments of the enrolled patients. For $t \ge \Delta + 1$, delayed outcomes are observed, so $\mathbf{H}_t$ continues to gather patient covariates and treatments and also includes delayed patient outcomes. For $t \ge T + 1$, the manager no longer enrolls patients, but delayed patient outcomes are included in $\mathbf{H}_t$ until time $T + \Delta$. Thus,

$$\mathbf{H}_t := \begin{cases} \emptyset, & \text{for } t = 0 \\ (\mathbf{X}_1, \ldots \mathbf{X}_t, W_1, \ldots, W_t), & \text{for } 1 \le t \le \Delta \\ (\mathbf{X}_1, \ldots \mathbf{X}_t, W_1, \ldots, W_t, Y_1, \ldots, Y_{t-\Delta}), & \text{for } \Delta + 1 \le t \le T \\ (\mathbf{X}_1, \ldots \mathbf{X}_T, W_1, \ldots, W_T, Y_1, \ldots, Y_{t-\Delta}), & \text{for } T + 1 \le t \le T + \Delta. \end{cases} \tag{1}$$

An *allocation policy* $\pi = (\pi_t)_{t=1,2,\ldots,T}$ is a sequence of functions that map the available data, $\mathbf{H}_{t-1}$, to a probability distribution over the set of treatments for each of the possible covariate vectors; that is, $\pi_t(w \mid \mathbf{h}, \mathbf{x}) = \mathbb{P}(W_t = w \mid \mathbf{H}_{t-1} = \mathbf{h}, \mathbf{X}_t = \mathbf{x})$ is the probability that allocation policy π assigns treatment w to a patient with covariates $\mathbf{x}$ at time t given data $\mathbf{h}$. Thus, an *action* is not a treatment but rather a probability distribution over the available treatments. The allocation policy assigns a treatment after observing the covariate vector and is specified for all possible patient covariate vectors. Because the allocation policy only uses past information, it is a member of the set of nonanticipatory policies Π.

A *treatment strategy* is a mapping from the space of covariates to the set of treatments. We denote the space of all treatment strategies by $\mathbf{f} = \{f : \mathscr{X} \to \mathscr{W}\}$. The *implementation decision* $\tilde{f} \in \mathbf{f}$ is the treatment strategy implemented for future patients: $\tilde{W}_i = \tilde{f}(\tilde{\mathbf{X}}_i)$.

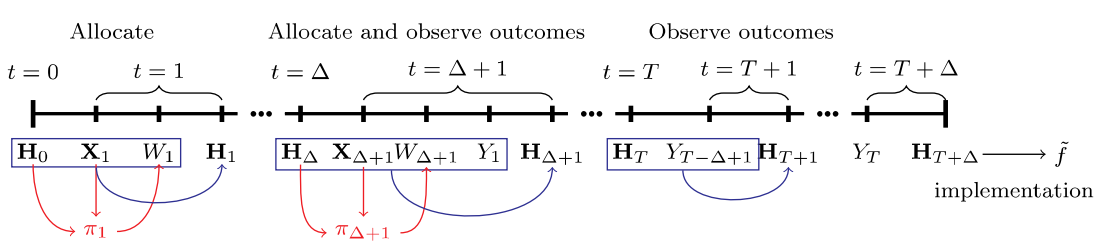
We denote the probability measure induced by an allocation policy π by $\mathbb{P}^\pi$, and the expectation with respect to that measure by $\mathbb{E}^\pi$. When the allocation policy is irrelevant for the event under consideration, we omit it from the superscript; for example, $\mathbb{E}[Y_t \mid \boldsymbol{\mu}, \mathbf{X}_t, W_t]$ is conditioned on the choice of treatment, so the policy is irrelevant, whereas $\mathbb{E}^\pi[Y_t]$ depends on allocation policy through the assigned treatment.

2.2. Trial Value

In our main model, we define the *trial value* with allocation policy π as the expected cumulative outcomes of the post-trial population of patients:

$$V^\pi = \mathbb{E}^\pi\left[\sum_{i=1}^{P} r_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_i, \tilde{f}(\tilde{\mathbf{X}}_i))\right] = P\mathbb{E}^\pi[r_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_1, \tilde{f}(\tilde{\mathbf{X}}_1))]. \tag{2}$$

Figure 1. (Color online) Timeline of Decisions and Information Acquisition



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The expectation is over the prior distribution of $\boldsymbol{\mu}$ (Section 2.5), the distributions of covariates of in-trial and post-trial patients (Section 2.4), and the treatments allocated during the trial, which may be randomized by allocation policy π.

We assume the implementation decision is the treatment strategy that maximizes the value for future patients given the information collected from the trial:

$$\tilde{f}(\mathbf{x}) = \arg\max_{\tilde{w}\in\mathscr{W}} \mathbb{E}[r_{\boldsymbol{\mu}}(\mathbf{x},\tilde{w})|\mathbf{H}_{T+\Delta}] \quad \forall \mathbf{x}\in\mathscr{X}.$$

Our objective is to find an allocation policy that maximizes trial value: $\sup_\pi V^\pi$. Because P is assumed fixed for now, it is sufficient to solve for the case $P=1$, and we do so here. Sampling to find the best treatment strategy at the end of information collection optimizes so-called offline rewards. We extend this model to include online rewards in Section 6.

2.3. Linear Response Function and Labeling of Active Regressors

In general, a response function $r_{\boldsymbol{\mu}}(\mathbf{x},w)$ maps a vector of covariates $\mathbf{x}\in\mathscr{X}$ and a treatment $w\in\mathscr{W}$ to the expected outcome. Here, we assume a linear model for $r_{\boldsymbol{\mu}}(\mathbf{x},w)$ to describe the effects of covariates and their potential interactions with treatments.

2.3.1. Preliminary Model with All Regressors. Let $\mu_{i,l}$ for $i\in\{0,1,\ldots,n\}$ and $l\in\{0,1,\ldots,m\}$ be the linear coefficient associated with treatment i and covariate l. Here, $\mu_{0,l}$ for $l=1,2,\ldots,m$ are the coefficients that represent effects of covariates regardless of treatment; $\mu_{i,0}$ for $i=1,2,\ldots,n$ are the coefficients for treatment effects that influence mean outcomes independent of covariates. This motivates our preliminary linear response function:

$$r_{\boldsymbol{\mu}}(\mathbf{x},w) = \underbrace{\mu_{0,0}}_{\text{intercept term}} + \sum_{i=1}^{n} \underbrace{\mathbb{1}_{w=i}\mu_{i,0}}_{\text{treatment effect}} + \sum_{l=1}^{m} \underbrace{x_l\mu_{0,l}}_{\text{prognostic term}} + \sum_{i=1}^{n}\sum_{l=1}^{m} \underbrace{\mathbb{1}_{w=i}x_l\mu_{i,l}}_{\text{predictive term}}, \tag{3}$$

where $\mathbb{1}$ is the indicator function. If the coefficient $\mu_{0,l}$ is nonzero, we say that covariate l is *prognostic*. If the coefficient $\mu_{i,l}$ is nonzero, we say that covariate l is *predictive* with respect to treatment i. If $\mu_{i,l}=0$ for all $i=0,1,\ldots,n$, then covariate l does not change the mean, and we say that covariate l is *idle*.

2.3.2. Active Regressors. In applications, contextual knowledge may suggest that some of these coefficients are zero (e.g., if the trial team believes that some covariates are prognostic but not predictive with respect to any treatment). Also, the model in (3) is overparameterized and thus not identifiable (e.g., it includes both intercept and all treatment effect terms).

We therefore consider response functions where some coefficients are constrained to be zero. Let the *labels* $\xi_{i,l}$ represent whether each coefficient is allowed to be nonzero: if $\xi_{i,l}=0$, then $\mu_{i,l}=0$; if $\xi_{i,l}=1$, then $\mu_{i,l}$ is free, needs to be estimated, and is referred to as *potentially active*. We define the labeling $\boldsymbol{\xi}$ to be the matrix of labels. The set of indices of potentially active coefficients is denoted $\Xi := \{(i,l) : \xi_{i,l}=1\}$.

Our *base model* for the response function for a fixed $\boldsymbol{\xi}$ is

$$r_{\boldsymbol{\mu}}(\mathbf{x},w) = \mu_{0,0}\xi_{0,0} + \sum_{i=1}^{n}\mathbb{1}_{w=i}\mu_{i,0}\xi_{i,0} + \sum_{l=1}^{m}x_l\mu_{0,l}\xi_{0,l} + \sum_{i=1}^{n}\sum_{l=1}^{m}\mathbb{1}_{w=i}x_l\mu_{i,l}\xi_{i,l}. \tag{4}$$

We now introduce operators that will be useful to simplify Expression (4) in the sequel using matrix multiplication. We let $\boldsymbol{\mu}$ be a $(n+1)(m+1)$-dimensional column vector of the regression coefficients as follows:

$$\boldsymbol{\mu} = \Big(\underbrace{\mu_{0,0},\mu_{0,1},\ldots,\mu_{0,m}}_{\text{associated with no treatment}}, \underbrace{\mu_{1,0},\ldots,\mu_{1,m}}_{\text{associated with treatment 1}}, \ldots, \underbrace{\mu_{n,0},\ldots,\mu_{n,m}}_{\text{associated with treatment } n} \Big)^\top. \tag{5}$$

Let $\boldsymbol{\mu}_{\boldsymbol{\xi}}\in\mathbb{R}^{|\Xi|}$ be the vector containing only potentially active coefficients in $\boldsymbol{\mu}$, as indicated by the labeling $\boldsymbol{\xi}$, and let $(w\otimes_{\boldsymbol{\xi}}\mathbf{x})\in\mathbb{R}^{1\times|\Xi|}$ be the row vector of active regressors such that

$$r_{\boldsymbol{\mu}}(\mathbf{x},w) = (w\otimes_{\boldsymbol{\xi}}\mathbf{x})\boldsymbol{\mu}_{\boldsymbol{\xi}}. \tag{6}$$

When we fix a known labeling and it is clear that we refer to it, we drop the $\boldsymbol{\xi}$ subscripts and use notation $r_{\boldsymbol{\mu}}(\mathbf{x},w) = (w\otimes\mathbf{x})\boldsymbol{\mu}$ instead of $r_{\boldsymbol{\mu}}(\mathbf{x},w) = (w\otimes_{\boldsymbol{\xi}}\mathbf{x})\boldsymbol{\mu}_{\boldsymbol{\xi}}$, where, with a slight abuse of notation, $\boldsymbol{\mu}\in\mathbb{R}^{|\Xi|}$ only includes potentially active coefficients. We allow for an unknown labeling of covariates in Section 9. We next give an example of a labeling.

Example 1 (Two Patient Types, Two Treatments). Consider a population with two types of patients, A and B, such that $\mathbf{X}_t=0$ for type A and $\mathbf{X}_t=1$ for type B. Consider two treatments, $\mathscr{W}=\{1,2\}$. Let $\xi_{i,l}=1$ for $i=1,2$ and $l=0,1$; and $\xi_{i,l}=0$ for $i=0$ and $l=0,1$; that is, the treatment effect and predictive terms are potentially active, but the intercept and prognostic terms are not. Then, $|\Xi|=4$, $\boldsymbol{\mu}=(\mu_{1,0},\mu_{1,1},\mu_{2,0},\mu_{2,1})^\top$, and

$$(1\otimes 0)\boldsymbol{\mu} = (1,0,0,0)\boldsymbol{\mu} = \mu_{1,0}, \quad (1\otimes 1)\boldsymbol{\mu} = (1,1,0,0)\boldsymbol{\mu} = \mu_{1,0}+\mu_{1,1},$$
$$(2\otimes 0)\boldsymbol{\mu} = (0,0,1,0)\boldsymbol{\mu} = \mu_{2,0}, \quad (2\otimes 1)\boldsymbol{\mu} = (0,0,1,1)\boldsymbol{\mu} = \mu_{2,0}+\mu_{2,1}.$$

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2.4. Distribution of Covariates

We assume the covariates of patients enrolled in the trial are exogenous, independent, and identically distributed (i.i.d.) random variables from a known distribution F_x: $\mathbf{X}_t \overset{i.i.d.}{\sim} F_x$. The covariates of post-trial patients are exogenous with a known distribution, $\tilde{\mathbf{X}}_i \overset{i.i.d.}{\sim} F_{\tilde{x}}$, which is not necessarily the same distribution as that of the trial patients. Allowing the distribution of covariates for patients enrolled in the trial to differ from that for patients treated post-trial provides useful flexibility, for example, to account for trial inclusion-exclusion criteria, different disease mix near the trial centers versus in the general population, and so on.

We impose two assumptions on the distributions of covariates F_x and $F_{\tilde{x}}$. Assumption 1 requires that $\mathbb{E}[\mathbf{X}_t^\top \mathbf{X}_t]$ and $\mathbb{E}[\tilde{\mathbf{X}}_t^\top \tilde{\mathbf{X}}_t]$ are bounded. This is similar to the standard assumption in regression analysis that $\mathbf{X}^\top\mathbf{X}$ is invertible, which ensures estimation can be done.

Assumption 1 (Integrability). *The distributions F_x of $\mathbf{X}_t$ and $F_{\tilde{x}}$ of $\tilde{\mathbf{X}}_i$ are such that $\mathbb{E}[\mathbf{X}_t^\top \mathbf{X}_t] < \infty$ and $\mathbb{E}[\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_i] < \infty$.*

Assumption 2 says that post-trial patients are represented in the trial with nonzero probability. Technically, it helps ensure convergence to the optimal treatment strategy. Practically, Assumption 2 should be considered along with exclusion criteria of the trial.

Assumption 2. *$F_{\tilde{x}}$ is absolutely continuous with respect to F_x.*

2.5. Bayesian Modeling and Updating

For a known labeling $\boldsymbol{\xi}$, we assume a conjugate normal model for patient outcomes and response function parameters and use the operator $\otimes$ of Section 2.3 in describing it.

Assumption 3 (Conjugate Normal Model). *We assume the following model for patient outcomes and for the parameters of the response function:*

$$Y_t \,|\, \boldsymbol{\mu}, \mathbf{X}_t, W_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}((W_t \otimes \mathbf{X}_t)\boldsymbol{\mu}, \sigma^2)$$
$$\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\theta}_0, \Sigma_0).$$

The parameters $\sigma^2 \in \mathbb{R}_{>0}$, $\boldsymbol{\theta}_0 \in \mathbb{R}^{|\Xi|}$, and $\Sigma_0 \in \mathbb{R}^{|\Xi|\times|\Xi|}$ (symmetric positive semidefinite) are known and bounded, where $\Xi := \{(i,l) : \xi_{i,l} = 1\}$.

With this model, the distribution of the unknown coefficients remains normally distributed after updating with Bayes' rule (Gelman et al. 2013, chapter 14). Specifically,

$$\boldsymbol{\mu} \,|\, \mathbf{H}_t \sim \mathcal{N}(\boldsymbol{\theta}_t, \Sigma_t),$$

where $\boldsymbol{\theta}_t$ and Σ_t are updated after each observation. For $t \le \Delta$, we do not observe any outcomes and do not update, so $\boldsymbol{\theta}_t = \boldsymbol{\theta}_0$ and $\Sigma_t = \Sigma_0$. For $t \ge \Delta + 1$, we use Bayes' rule and update with the following recursive equations:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \frac{Y_{t-\Delta} - (W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})\boldsymbol{\theta}_{t-1}}{\sigma^2 + (W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})\Sigma_{t-1}(W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})^\top} \Sigma_{t-1}(W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})^\top, \tag{7a}$$

$$\Sigma_t = \Sigma_{t-1} - \frac{\Sigma_{t-1}(W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})^\top (W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})\Sigma_{t-1}}{\sigma^2 + (W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})\Sigma_{t-1}(W_{t-\Delta} \otimes \mathbf{X}_{t-\Delta})^\top}. \tag{7b}$$

We discuss specifying the prior distribution in Section 7 and provide more details on the Bayesian updating equations in Online Appendix EC.1.

We define a *pipeline state* $\mathbf{J}_t$ to contain the covariates and treatments of patients that have been assigned to treatment, but whose outcomes have not been observed at time t:

$$\mathbf{J}_t := \begin{cases} \emptyset, & \text{for } \Delta = 0 \text{ or } t = 0 \text{ or } t = T + \Delta \\ (\mathbf{X}_1, W_1, \ldots, \mathbf{X}_t, W_t), & \text{for } 1 \le t \le \Delta \\ (\mathbf{X}_{t-\Delta+1}, W_{t-\Delta+1}, \ldots, \mathbf{X}_t, W_t), & \text{for } \Delta + 1 \le t \le T \\ (\mathbf{X}_{t-\Delta+1}, W_{t-\Delta+1}, \ldots, \mathbf{X}_T, W_T), & \text{for } T + 1 \le t \le T + \Delta - 1. \end{cases} \tag{8}$$

We denote the knowledge state by $\mathbf{K}_t := (\boldsymbol{\theta}_t, \Sigma_t, \mathbf{J}_t)$. Note that $\mathbf{K}_t$ is a sufficient statistic for the unknown model parameters. Let τ_t be the transition function that updates the knowledge state recursively, with $\mathbf{K}_t = \tau_t(\mathbf{K}_{t-1}, \mathbf{X}_t, W_t, Y_{t-\Delta})$ for $t = 1, 2, \ldots, T + \Delta$, respecting Bayes' rule in (7) and the definition of the pipeline state. The function τ_t has a subscript t to handle the cases $t \le \Delta$ and $t \ge T + 1$ where $\mathbf{X}_t$, W_t, or $Y_{t-\Delta}$ may not exist.

3. Optimal Allocation Policy

In this section, we describe the optimal policy. We first describe the implementation decision that maximizes trial value given the knowledge state after all data from the trial has been observed, at time $T + \Delta$. We then present a dynamic program for solving for the optimal allocation policy of patients to arms as a function of their covariates and show the existence of an optimal solution. The dynamic program is generally computationally infeasible to solve yet provides the foundation for the development of expected value of information heuristics (Section 1.2) and their theoretical properties (Section 5).

3.1. Implementation Decision

Recall our linear model, $r_{\boldsymbol{\mu}}(\mathbf{x}, w) = (w \otimes \mathbf{x})\boldsymbol{\mu}$, which has conditional mean $\mathbb{E}[r_{\boldsymbol{\mu}}(\mathbf{x}, w) \,|\, \mathbf{K}_t] = (w \otimes \mathbf{x})\boldsymbol{\theta}_t$ given knowledge at time t. The treatment that optimizes outcomes for a new patient with covariates $\mathbf{x}$ given knowledge at time t is

$$\tilde{f}_{\boldsymbol{\theta}_t}(\mathbf{x}) = \arg\max_{w \in \mathcal{W}} (w \otimes \mathbf{x})\boldsymbol{\theta}_t, \tag{9}$$

where the notation $\tilde{f}_{\boldsymbol{\theta}_t}$ emphasizes the dependence of

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the treatment strategy on $\boldsymbol{\theta}_t$, and not all of $\mathbf{H}_t$ or $\mathbf{K}_t$. The implementation decision given all trial data $\mathbf{H}_{T+\Delta}$ is therefore

$$\tilde{f}(\mathbf{x}) = \tilde{f}_{\boldsymbol{\theta}_{T+\Delta}}(\mathbf{x}) = \arg\max_{w \in \mathscr{W}} (w \otimes \mathbf{x}) \boldsymbol{\theta}_{T+\Delta}. \tag{10}$$

3.2. Dynamic Program

We now write $\sup_{\pi \in \Pi} V^\pi$ as a dynamic program. Let $\mathscr{K}_t$ be the set of feasible knowledge states at time $t \in \{0, 1, \ldots, T\}$, and let $\mathbf{k} = (\boldsymbol{\theta}, \Sigma, \mathbf{j}) \in \mathscr{K}_t$ be a knowledge state at time t with posterior mean $\boldsymbol{\theta}$ and covariance matrix Σ for $\boldsymbol{\mu}$ and pipeline state $\mathbf{j}$.

By construction, the terminal reward $G(\mathbf{k})$ for knowledge states $\mathbf{k} \in \mathscr{K}_T$ that are feasible at time T, using the implementation decision in (10), is

$$\begin{aligned} G(\mathbf{k}) &= \mathbb{E}[(\tilde{f}_{\boldsymbol{\theta}_{T+\Delta}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\theta}_{T+\Delta} | \mathbf{K}_T = \mathbf{k}] \\ &= \mathbb{E}\left[\max_{\tilde{w} \in \mathscr{W}} (\tilde{w} \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\theta}_{T+\Delta} | \mathbf{K}_T = \mathbf{k}\right], \end{aligned} \tag{11}$$

with the expectation taken over the outcomes of the pipeline patients that are to be observed by time $T + \Delta$ and that help determine the optimal treatment strategy. Also by construction, the value-to-go $V_t^\pi(\mathbf{k})$ for an allocation policy π (not necessarily the optimal allocation policy), evaluated at time $t \le T - 1$ with state $\mathbf{k} \in \mathscr{K}_t$, is

$$V_t^\pi(\mathbf{k}) = \mathbb{E}^{\pi_{t+1}}[V_{t+1}^\pi(\tau_{t+1}(\mathbf{k}, \mathbf{X}_{t+1}, W_{t+1}, Y_{t-\Delta+1})) | \mathbf{K}_t = \mathbf{k}], \quad \text{for } t = 0, \ldots, T-1, \tag{12}$$

where π_t determines the treatment W_t, and τ_t updates the knowledge state as in Section 2.5. Thus, the optimal value-to-go upon treating patient t is given by the dynamic program:

$$\begin{aligned} V_t(\mathbf{k}) &= \sup_{\pi_{t+1}} \mathbb{E}^{\pi_{t+1}}[V_{t+1}(\tau_{t+1}(\mathbf{k}, \mathbf{X}_{t+1}, W_{t+1}, Y_{t-\Delta+1})) | \mathbf{K}_t = \mathbf{k}], \quad \text{for } t = 0, 1, \ldots, T-1 \\ V_T(\mathbf{k}) &= G(\mathbf{k}). \end{aligned} \tag{13}$$

With this formulation, we can prove the existence of an optimal allocation policy π^* that solves $\max_{\pi \in \Pi} V^\pi$. The optimal allocation policy is deterministic (does not randomize treatments) and Markov (only depends on the knowledge state). Our proof of this result in Online Appendix EC.2 maps (13) to the model in Bertsekas and Shreve (1978, chapter 8).

Proposition 1. *The dynamic program in* (13) *satisfies that* $V_0(\mathbf{K}_0) = \sup_{\pi \in \Pi} V^\pi$, *and there exists a deterministic Markov allocation policy* π^*, *which can be computed using Bellman's recursion in* (13), *such that* $V_t(\mathbf{k}) = V_t^{\pi^*}(\mathbf{k})$ *for all* $\mathbf{k} \in \mathscr{K}_t$ *and all* $t = 0, \ldots, T$.

Because an optimal allocation policy is attainable, we use $V^* = \max_{\pi \in \Pi} V^\pi$ to denote the trial value attained by an optimal allocation policy π^*.

4. Heuristics for Learning with Predictive and Prognostic Covariates

The computation of the optimal allocation policy suffers from the curse of dimensionality. Thus, we develop heuristics for assigning patients to arms that are computationally tractable, perform well, and have useful analytic properties. In this section, we extend the expected value of information approach of Bayesian sequential optimization to account for predictive and prognostic covariates. We will derive theoretical properties of the heuristics in Section 5 and assess their performance numerically in Section 8.

4.1. Functional Expected Value of Information (*f*EVI)

We call this heuristic *f*EVI because it uses the EVI approach to learn a treatment strategy, a *function* from covariates to treatments. The *f*EVI allocation policy makes at each step the sampling decision that maximizes the expected value of information of one more sample.

In particular, the *f*EVI index, $\nu_t(\mathbf{x}, w)$, is the value gained from allocating one additional patient with covariates $\mathbf{x}$ to treatment w, and selecting a treatment strategy after waiting to observe the outcomes of that patient and the patients in the pipeline, over selecting a treatment strategy using only the outcomes of the patients that have been allocated already, including the ones in the pipeline (all conditional on $\mathbf{K}_t$). For $0 \le t \le T - 1$,

$$\begin{aligned} \nu_t(\mathbf{x}, w) &= \mathbb{E}[(\tilde{f}_{\boldsymbol{\theta}_{t+\Delta+1}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} | \mathbf{K}_t, \mathbf{X}_{t+1} = \mathbf{x}, W_{t+1} = w] \\ &\quad - \mathbb{E}[(\tilde{f}_{\boldsymbol{\theta}_{t+\Delta}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} | \mathbf{K}_t] \\ &= \underbrace{\mathbb{E}\left[\max_{\tilde{w} \in \mathscr{W}} (\tilde{w} \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\theta}_{t+\Delta+1} | \mathbf{K}_t, \mathbf{X}_{t+1} = \mathbf{x}, W_{t+1} = w\right]}_{\text{implement after one more patient and pipeline clears}} \\ &\quad - \underbrace{\mathbb{E}\left[\max_{\tilde{w} \in \mathscr{W}} (\tilde{w} \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\theta}_{t+\Delta} | \mathbf{K}_t\right]}_{\text{implement after pipeline clears}}. \end{aligned} \tag{14}$$

The *f*EVI allocation policy assigns the treatment with the highest index, $W_{t+1} = \arg\max_{w \in \mathscr{W}} \nu_t(\mathbf{X}_{t+1}, w)$, $0 \le t \le T - 1$, breaking ties by sampling uniformly at random between treatments with the largest index.

4.2. MC Estimates of *f*EVI and Randomization of Treatments

The computation of the *f*EVI index requires integration over a potentially high-dimensional space, and therefore quadrature can present challenges, particularly if $\Delta > 0$ or if there are multiple continuous-valued covariates. We thus propose an allocation

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policy that approximates the fEVI index using MC simulation.

MC simulation helps address the challenge of high-dimensional integration and also introduces some randomization in allocating treatments. Clinical trial practice may use randomization to arms to address sampling biases (Piantadosi 1997). We also present allocation policies that allow a trial manager to further control randomization to arms.

4.2.1. MC fEVI Allocation Policy (fEVI-MC). The expectations in (14) that determine the fEVI index can be estimated by simulating the unknown regression coefficients and the outcomes of pipeline patients; optimizing treatments conditional on that data; and estimating rewards for a simulated post-trial population. An advantage of this basic approach is that models that do not satisfy the conjugate normality of Assumption 3 can be handled.

If Assumption 3 is satisfied, as we assume here, we can improve computational efficiency in the simulation estimates of the fEVI-indices of (14). The fEVI-MC function in Algorithm 1 implements an MC estimator $\hat{v}_t(\mathbf{x}, w)$ of the indices $v_t(\mathbf{x}, w)$ in (14) for each w and presents several efficiency improvements relative to the basic approach above. There are three sources of randomness to simulate: (i) the outcomes of Δ'_t pipeline patients that are observed through time $t+\Delta$, where $\Delta'_t = \min\{t, \Delta\}$ is the size of the pipeline at time t; (ii) the covariates of post-trial patients to estimate mean rewards; and (iii) outcomes of patient $t+1$ with covariate $\mathbf{X}_{t+1}$ for each treatment. We observe that, conditional on sources of randomness (i) and (ii), the integration used to compute the (conditional) EVI over randomness from source (iii) can be computed exactly using a subroutine of the correlated knowledge gradient (cKG; Frazier et al. 2009). This is computed in steps 11–13, where $h(\mathbf{a}, \mathbf{b}) = \mathbb{E}[\max_i a_i + b_i Z] - \max_i a_i$, where $Z \sim \mathcal{N}(0,1)$, and a_i and b_i are the ith entries of vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. See Online Appendix EC.4 for justification. This enables a powerful variance reduction for estimating fEVI indices based on conditional expectations. For (i), we average such computations over η^{on} realizations of $\boldsymbol{\theta}_{t+\Delta}$ given $\mathbf{K}_t$ simulated in step 7 using standard Bayesian results for the preposterior distribution. For (ii), we simulate η^{off} post-trial patients in step 8. We use a second variance reduction, common random numbers (CRN) across treatment choices, for the simulation of (i) and (ii), to improve the ability to identify the largest index. Alban et al. (2024) further discuss how these variance reductions improve the selection of the largest fEVI index.

Algorithm 1 (fEVI-MC: Estimate fEVI Indices with Common Random Numbers Across Pipeline and Post-trial Patients and with Conditional MC)

1: **function** fEVI-MC$(t, \mathbf{K}_t, \mathbf{X}_{t+1}; \eta^{\text{on}}, \eta^{\text{off}})$
2: Let $\Delta'_t = \min\{t, \Delta\}$ ▷ Compute number of patients in pipeline.
3: $\hat{\boldsymbol{\sigma}} \leftarrow \tilde{\boldsymbol{\sigma}}(\Sigma_t, \mathbf{W}_{(t-\Delta'_t+1):(t)}, \mathbf{X}_{(t-\Delta'_t+1):(t)})$ ▷ Prepost std dev in (EC.5) for pipeline
4: Compute $\Sigma_{t+\Delta}$ ▷ Compute post var, after pipeline clears (7b)
5: **for** j in $\{1, \ldots, \eta^{\text{on}}\}$ **do** ▷ For each realization of pipeline outcomes …
6: $\hat{\mathbf{Z}} \sim \mathcal{N}(0, I_{\Delta'_t})$ ▷ Simulate perturbation noise for posterior, with pipeline of length Δ'_t
7: $\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}_t + \hat{\boldsymbol{\sigma}}\hat{\mathbf{Z}}$ ▷ Simulate a posterior mean $\boldsymbol{\theta}_{t+\Delta}$ when pipeline clears
8: $\hat{\mathbf{X}}_{1:\eta^{\text{off}}} \overset{i.i.d.}{\sim} F_{\tilde{x}}$ ▷ Generate covariates for η^{off} post-trial patients
9: **for all** $w \in \mathscr{W}$ **do** ▷ For each treatment that could be given to patient type $\mathbf{X}_{t+1}$ …
10: **for all** $i = 1, 2, \ldots, \eta^{\text{off}}$ **do** ▷ For each post-trial patient …
11: $\mathbf{a} \leftarrow ((1, 2, \ldots, n) \otimes \hat{\mathbf{X}}_i)\hat{\boldsymbol{\theta}}$ ▷ Vector of means for n treatments, type $\hat{\mathbf{X}}_i$ patients
12: $\mathbf{b} \leftarrow ((1, 2, \ldots, n) \otimes \hat{\mathbf{X}}_i)\tilde{\boldsymbol{\sigma}}(\Sigma_{t+\Delta}, w, \mathbf{X}_{t+1})$ ▷ Prepost std dev (EC.5) if patient $t+1$ has covariates $\mathbf{X}_{t+1}$ with treatment $W_{t+1} = w$
13: $\log(v_{w,j,i}) \leftarrow \log(h(\mathbf{a}, \mathbf{b}))$ ▷ Conditional EVI, where $\log(h(\mathbf{a}, \mathbf{b}))$ is the v …
14: **end for** ▷ … for cKG computed in algorithm 2 in Frazier et al. (2009).
15: $\log(\hat{v}_{w,j}) \leftarrow \log((1/\eta^{\text{off}})\sum_{i=1}^{\eta^{\text{off}}} v_{w,j,i})$ ▷ Estimate conditional EVI for w, given $\hat{\boldsymbol{\theta}}$
16: **end for**
17: **end for**
18: **for all** $w \in \mathscr{W}$ **do**
19: $\log(\hat{v}_t(\mathbf{X}_{t+1}, w)) \leftarrow \log((1/\eta^{\text{on}})\sum_{j=1}^{\eta^{\text{on}}} \hat{v}_{w,j})$ ▷ fEVI-MC-index for w, ave. over …
20: **end for** ▷ … pipeline and post-trial patients
21: **return** $W_{t+1} = \arg\max_{w \in \mathscr{W}} \log(\hat{v}_t(\mathbf{X}_{t+1}, w))$ ▷ Treatment with largest fEVI-MC-index
22: **end function** ▷ … (break ties uniformly at random)

Online Appendix EC.1 derives the so-called preposterior standard deviation used in fEVI-MC, $\tilde{\boldsymbol{\sigma}}(\Sigma_{t_1}, \mathbf{W}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)}, \mathbf{X}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)})$, a $|\Xi| \times ((t_2 - t_1) - (\Delta - \Delta'_{t_1}))$ matrix such that $\tilde{\boldsymbol{\sigma}}(\Sigma_{t_1}, \mathbf{W}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)}, \mathbf{X}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)})$ $\tilde{\boldsymbol{\sigma}}^{\top}(\Sigma_{t_1}, \mathbf{W}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)}, \mathbf{X}_{(t_1-\Delta'_{t_1}+1):(t_2-\Delta)})$ is the covariance of the mean $\boldsymbol{\theta}_{t_2}$, given Σ_{t_1}, and the outcomes of pipeline patients with indices $t_1 - \Delta'_{t_1} + 1, t_1 - \Delta'_{t_1} + 2, \ldots, t_2 - \Delta$ to be observed by time t_2. (Recall that $\Delta'_{t_1} = \min\{t_1, \Delta\}$.)

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Step 3 uses $t_1 = t$ and $t_2 = t + \Delta$. Step 12 uses $t_1 = t + \Delta$ and $t_2 = t + \Delta + 1$.

4.2.2. Randomized *f* EVI Allocation Policies: *f* EVI-rand and *f* EVI-MC-Rand. The *f* EVI allocation policy is deterministic. Although *f* EVI-MC allows for some randomization, the degree of randomness in *f* EVI-MC is not under explicit control of the trial manager. To accommodate greater control over randomization, allocation policy *f* EVI-rand samples a treatment uniformly at random with probability $\epsilon > 0$, and chooses the treatment with the largest *f* EVI index with probability $1 - \epsilon$ (an approach followed by Cheng and Berry (2007), Lai et al. (2013), and Williamson et al. (2017)). When the *f* EVI indices are estimated using the *f* EVI-MC algorithm, we call the policy *f* EVI-MC-rand.

5. Theoretical Properties of Expected Value of Information Heuristics

This section discusses the theoretical properties of the expected value of information heuristics of Section 4. EVI heuristics are designed to be one-step look-ahead optimal, at least when computed exactly. Although they are myopic, EVI heuristics may also be optimal as the sample size goes to infinity. These properties, in the small and large sample size regimes, were already proved for settings where there are no covariates (Frazier et al. 2009, Chick et al. 2022) or where covariates are selected by the allocation policy (Ding et al. 2021). Here, we extend these properties for the setting where the covariates are random arrivals, satisfying Assumptions 1–3 and for a positive fixed delay in observing outcomes. Proofs of claims in this section are found in Online Appendix EC.3.

If the sample size is $T = 1$, the *f* EVI allocation policy is optimal for (13).

Proposition 2. *If $T = 1$, then $V^{f\,\mathrm{EVI}}(\mathbf{K}_0) = V^*(\mathbf{K}_0)$.*

The *f* EVI-MC, *f* EVI-rand, and *f* EVI-MC-rand allocation policies are not necessarily optimal for $T = 1$, due to their reliance on MC samples, randomization, or both.

For asymptotically large T, each of *f* EVI, *f* EVI-rand, and *f* EVI-MC-rand can be shown to be asymptotically optimal. To show this, we first show an upper bound on the value function and then show that the value approaches this upper bound as the sample size grows without bound. Because the following results characterize V^π as a function of the sample size, we use $V^\pi(\mathbf{K}_0; T)$ to make the dependence on $\mathbf{K}_0$ and T explicit.

Let $U(\boldsymbol{\theta}_0, \Sigma_0) := \mathbb{E}[\max_{w \in \mathscr{W}} (w \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \boldsymbol{\theta}_0, \Sigma_0]$ be the expected rewards obtained by an *oracle* that knows the parameters exactly ex ante. We first provide a characterization of the estimated parameters and the value function.

Theorem 1. *For a given allocation policy π, there exists a random vector $\boldsymbol{\theta}^\pi_\infty$ and a random matrix Σ^π_∞ such that $\boldsymbol{\theta}_t \to \boldsymbol{\theta}^\pi_\infty$ almost surely and $\Sigma_t \to \Sigma^\pi_\infty$ almost surely. Moreover, $V^\pi(\mathbf{K}_0; T)$ is bounded above by $U(\boldsymbol{\theta}_0, \Sigma_0)$, and $V^\pi(\mathbf{K}_0; \infty) := \lim_{T\to\infty} V^\pi(\mathbf{K}_0; T)$ exists.*

Given the upper bound on the trial value, we define an asymptotically optimal allocation policy to be one that achieves the upper bound as the sample size goes to infinity.

Definition 1. If π is such that $V^\pi(\mathbf{K}_0; \infty) = U(\boldsymbol{\theta}_0, \Sigma_0)$, then we say that π is an *asymptotically optimal allocation policy*.

Theorem 2 states that allocation policies that have a positive probability of sampling the treatment with the largest expected value of information are asymptotically optimal.

Theorem 2. *Let allocation policy π be such that, for $0 \le t \le T - 1$, if $v_t(\mathbf{x}, w) \ge v_t(\mathbf{x}, v)$ $\forall v \in \mathscr{W}$, then there exists a $\delta > 0$ so that $\mathbb{P}^\pi(W_{t+1} = w \,|\, \mathbf{K}_t, \mathbf{X}_{t+1} = \mathbf{x}) \ge \delta$. Then, π is asymptotically optimal.*

The proof of Theorem 2 shows that policies are asymptotically optimal if they always sample, with probability bounded away from zero, covariate-treatment combinations with positive value of information. Previous proofs of asymptotic optimality in settings without covariates (Frazier et al. 2009, Xie et al. 2016) show that policies are asymptotically optimal if they sample all treatments infinitely often (excluding treatments fully correlated with other treatments). The proof of Ding et al. (2021), which shows asymptotic optimality in the presence of covariates when the covariates are a choice of the allocation policy, also relies on showing that each treatment is sampled infinitely often.[1] An alternative approach is necessary in the presence of randomly arriving covariates because sampling each alternative infinitely often is not enough to achieve asymptotic optimality.[2] Using Theorem 2, we show that *f* EVI, *f* EVI-rand, and *f* EVI-MC-rand are asymptotically optimal.

Theorem 3. *Allocation policies* *f* EVI, *f* EVI-rand, *and* *f* EVI-MC-rand *are asymptotically optimal.*

Our proof of Theorem 1 (Corollary EC.3 in Online Appendix EC.3.3.1) shows that $V^\pi(\mathbf{K}_0; \infty) = \mathbb{E}^\pi[(\tilde{f}_{\boldsymbol{\theta}^\pi_\infty}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \mathbf{K}_0]$ for any allocation policy π. The definitions of U and $\tilde{f}_{\boldsymbol{\mu}}$ imply that $U(\boldsymbol{\theta}_0, \Sigma_0) = \mathbb{E}[(\tilde{f}_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \boldsymbol{\theta}_0, \Sigma_0]$. For an asymptotically optimal π we thus have

$$\mathbb{E}^\pi[(\tilde{f}_{\boldsymbol{\theta}^\pi_\infty}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \mathbf{K}_0] = \mathbb{E}[(\tilde{f}_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \mathbf{K}_0].$$

By the definition of $\tilde{f}_{\boldsymbol{\mu}}$, we know that $(\tilde{f}_{\boldsymbol{\theta}^\pi_\infty}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \le (\tilde{f}_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu}$, which implies

$$(\tilde{f}_{\boldsymbol{\theta}^\pi_\infty}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} = (\tilde{f}_{\boldsymbol{\mu}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \text{ a.s.}$$

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Thus, asymptotically optimal allocation policies (including, by Theorem 3, fEVI, fEVI-rand, and fEVI-MC-rand) result in convergence to an optimal treatment strategy. Although a uniform random allocation of treatments for each patient is also asymptotically optimal, numerical results in Section 8 suggest it is not as effective with small sample sizes.

We have not proven fEVI-MC to be asymptotically optimal in general, but prove its optimality for some interesting special cases (e.g., if the set of covariate values is finite and there is no delay) in Online Appendix EC.4.2. That appendix also explores other properties of fEVI-MC, and Section 8 will demonstrate its usefulness in practice.

6. Allocation Policies with Online Rewards

Our main model above maximizes offline rewards with a known labeling for covariate structure, where rewards accrue from post-trial patient outcomes. Alternatively, the modeler may wish to focus on the outcomes of patients *in the trial*, for example, when the total population of patients is small, as may be the case for rare diseases. This may be modeled with online rewards. Here, the online value is the expected cumulative reward from the outcomes of patients who receive treatment in the trial:

$$V^{\pi,\text{on}} = \mathbb{E}^{\pi}\left[\sum_{t=1}^{T} r_{\boldsymbol{\mu}}(\mathbf{X}_t, W_t)\right]. \tag{15}$$

Ryzhov et al. (2012) show how the expected value of information approach for offline rewards can be extended to handle online rewards in a noncontextual setting. We similarly extend our offline rewards fEVI approach to handle online rewards in the contextual learning setting of precision medicine in Online Appendix EC.5. We will assess policies that seek to optimize online rewards in numerical experiments in Section 8.5.

Remark 1. We use a superscript to qualify that a policy uses online rewards. For example, $f\text{EVI}^{\text{on}}$ is a variation of fEVI that seeks to maximize the expected online reward, $V^{\pi,\text{on}}$, in (15). The lack of a superscript ^{on} implies an offline reward objective.

Remark 2. We also define a combined reward, $V^{\pi,\text{on+off}} = V^{\pi,\text{on}} + PV^{\pi}$, which sums the expected online reward $V^{\pi,\text{on}}$ in (15) for T in-trial patients and the expected reward V^{π} for P post-trial patients from (2). The associated allocation policy is denoted $f\text{EVI}^{\text{on+off}}$. We show numerical experiments with that policy in Online Appendix EC.7.6.

7. Modeler's Prior Distribution and Distribution of Problem Instances

The modeler's beliefs about the distribution of problem instances is assumed to be specified by the conjugate normal prior distribution $\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\theta}_0, \Sigma_0)$ in Assumption 3. It is common to use a Gaussian process model to specify realizations of means (Frazier et al. 2009).

We distinguish between a modeler's prior distribution of problem instances, specified by $\boldsymbol{\theta}_0$ and Σ_0, and the distribution of problem instances that nature presents to the modeler. For the latter, we allow nature to generate problem instances $\boldsymbol{\mu}$ with a $\mathcal{N}(\boldsymbol{\theta}_0^{nat}, \Sigma_0^{nat})$ distribution, which we refer to as *nature's distribution*.

It may be useful to allow the modeler's prior distribution to differ from nature's distribution. For example, we may wish to assess the performance of allocation policies when nature chooses a specific problem structure (e.g., a *slippage configuration* where a single "best" treatment is precisely δ units better than any alternative treatment, independent of the covariates; Shen et al. 2021). We call this a *fixed instance* setting because we let $\boldsymbol{\theta}_0^{nat}$ be a fixed value of the true means $\boldsymbol{\mu}$ (and let Σ_0^{nat} be the zero matrix). We may also wish to allow that problem instances be randomly sampled to assess the average performance of an allocation policy (Branke et al. 2007). We call this a *random instance* setting because we allow nature to generate random draws of $\boldsymbol{\mu}$ from nature's distribution $\mathcal{N}(\boldsymbol{\theta}_0^{nat}, \Sigma_0^{nat})$.

In general, a modeler may wish to elicit a probability distribution for problem instances using known elicitation techniques (O'Hagan et al. 2006). If the elicitation is done well and Assumption 3 holds, then one expects that $\boldsymbol{\theta}_0 = \boldsymbol{\theta}_0^{nat}$ and $\Sigma_0 = \Sigma_0^{nat}$.

To reduce the dimensionality of the prior covariance matrix, we assume a Gaussian kernel (Frazier et al. 2009), although other kernels may be used (Chen et al. 2013):

$$\Sigma_{0,\ell_1,\ell_2} = \begin{cases} \sigma_0^2 & \text{if } \ell_1 = \ell_2 \text{ (same treatment and covariate)} \\ 0 & \text{if } \kappa_x(\ell_1) \neq \kappa_x(\ell_2) \text{ (different covariate)} \\ \sigma_0^2 e^{-\psi d(\kappa_w(\ell_1), \kappa_w(\ell_2))} & \text{otherwise (same covariate and different treatment),} \end{cases} \tag{16}$$

where σ_0^2 is the prior variance for the unknown value of coefficients $\boldsymbol{\mu}$, $\kappa_x(\ell)$ is the covariate index associated with the regression coefficient indexed by $\ell = 1, \ldots, |\Xi|$ (0 for intercept or treatment effect terms), $\kappa_w(\ell)$ is the treatment associated with the regression coefficient indexed by $\ell = 1, \ldots, |\Xi|$ (0 for intercept or prognostic terms), and d is a distance function between treatments, which is smaller for treatments that act with similar underlying mechanisms and

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larger if the treatments act with different mechanisms. If either argument of d is zero, corresponding to intercept or prognostic terms, we set d to be infinity (no correlation). Larger values of $\psi > 0$ model a lower correlation between similar treatments.

If pilot data are available, for example, from a Phase 2 clinical trial, then it is possible to use empirical Bayes methods to define the prior distribution (Gelman et al. 2013). Here, we use the linear least squares estimate to define the prior with empirical Bayes.

Related work for optimal sequential learning without covariates suggests that it may be useful to consider manipulations of the modeler's prior distribution to reduce the risk of selecting an inferior alternative (Powell and Ryzhov 2012, Chick et al. 2022). We adopt such a mindset here and define the *robust prior distribution* as follows, with an adaptation to account for the distinct nature of predictive and prognostic covariates:

- For coefficients with index ℓ associated with prognostic or intercept terms, we set $\theta^{rob}_{0,\ell} = 0$, to push for evidence collection to confirm that the coefficients are nonzero;
- For coefficients with index ℓ associated with predictive or treatment effect terms, we set $\theta^{rob}_{0,\ell}$ artificially high, to push sampling to confirm whether such effects are active or not, with a "fudge" factor z_α (two unless otherwise specified) times a standard error:

$$\theta^{rob}_{0,\ell} = \max_{\ell':\kappa_x(\ell')=\kappa_x(\ell)} (\theta_{0,\ell'}) + z_\alpha \cdot \sqrt{\max_{\ell':\kappa_x(\ell')=\kappa_x(\ell)} (\Sigma_{0,\ell',\ell'})}; \quad (17)$$

- We set $\Sigma^{rob}_0 = c \cdot \Sigma_0$, for some $c \geq 1$, to reduce the effect of the prior distribution on the inference and push for further exploration. We let $c = 4$ unless otherwise specified.

8. Empirical Performance of Expected Value of Information Heuristics

The analytical results in Section 5 provide optimality guarantees for asymptotic sample sizes or a sample size of one. We now assess empirically the learning efficiency of our EVI-based heuristics for intermediate sample sizes. We describe performance metrics and comparator heuristics in Section 8.1. The testbed consists of two applications that are described in Section 8.2 and are motivated by the treatment of sepsis.

We address the following questions in this section: If the correct labeling is known, does the EVI approach learn treatment strategies more efficiently than other approaches (Section 8.3)? How useful is MC simulation for estimating EVI indices in the presence of delays and continuous covariates (Section 8.4)? What is the efficiency of the proposed EVI-based approach for online rewards, assuming knowledge of the correct labeling (Section 8.5)? Does randomization, which is important in implementing clinical trials, significantly degrade the performance of the proposed EVI-based algorithms (Section 8.6)?

8.1. Performance Metrics and Comparator Allocation Policies

We measure the ability of allocation policies to learn the best treatment strategy for post-trial patients with the *expected opportunity cost* (EOC). EOC is the expected difference between the value obtained by an oracle and by a policy π for a given set of coefficients $\boldsymbol{\mu}$:

$$\mathrm{EOC}^\pi(\boldsymbol{\mu}; T) := \mathbb{E}^\pi\left[\max_{w\in\mathscr{W}} (w \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} - (\tilde{f}_{\boldsymbol{\theta}_{T+\Delta}}(\tilde{\mathbf{X}}_1) \otimes \tilde{\mathbf{X}}_1)\boldsymbol{\mu} \,|\, \boldsymbol{\mu}\right]. \quad (18)$$

EOC averages over trial outcomes through π and over the covariates of post-trial patients $\tilde{\mathbf{X}}_1$ after the treatment strategy $\tilde{f}$ is adopted. Its values are conditional on the true means, $\boldsymbol{\mu}$, of a given problem instance. For a random instance setting, we average realizations of random problem instances, that is, $\mathrm{EOC}^\pi(T) = \mathbb{E}[\mathrm{EOC}^\pi(\boldsymbol{\mu}; T) \,|\, \boldsymbol{\theta}^{nat}_0, \Sigma^{nat}_0]$. Better allocation policies obtain lower values of EOC. We present results for two other metrics: the cumulative regret for online rewards in Section 8.5 and the probability of incorrect selection (PICS) for best-arm identification in Online Appendix EC.7.2.

We compare our EVI-based allocation policies against several other policies for assigning treatments to the next patient in the trial. The Thompson sampling (TS) allocation policy draws a sample of $\boldsymbol{\mu}$ from its posterior distribution and then assigns the best treatment, in expectation, given that $\boldsymbol{\mu}$ (Thompson 1933). *Top-two Thompson sampling* (TTTS) is an adaptation of Thompson sampling (Russo 2020): With probability β, the policy assigns the treatment assigned by the TS allocation policy, and with probability $1-\beta$, the policy resamples $\boldsymbol{\mu}$ from the posterior distribution until a sample is drawn such that the best treatment differs from that recommended by TS. We use $\beta = 0.5$. The *Random* policy assigns treatments uniformly at random, regardless of covariates. The *Biased Coin* (BC*) policy balances the values of prognostic covariates across arms for a given set of predictive covariate values, in the spirit of Pocock and Simon (1975). Initial work on the contextual optimal computing budget allocation (OCBA; Cakmak et al. 2021) assumes that a decision maker can choose both the treatment and covariate values of the next patient to be enrolled to improve the probability of correctly selecting the best treatment for that patient. We use an adaptation of OCBA suited to our context, where covariates are randomly determined by patient arrivals. The response-adaptive biased coin (RABC), motivated by Zhao et al. (2022), combines the BC* and OCBA policies for balancing covariates and treatments. The structure of these BC*, OCBA, and RABC

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allocation policies assumes that a covariate is predictive or prognostic (but not both) and that if a covariate is predictive with respect to one treatment, it is assumed predictive with respect to all treatments. Online Appendix EC.6 gives details and further discussion.

The fEVI policy and its variations in Section 4 are *exploration* focused: They seek to maximize offline rewards. The most direct comparators that tend to explore are the BC*, OCBA, and RABC policies. Policies that tend to *balance exploration and exploitation* to maximize online, or a combination of online and offline, rewards include TS and our policies with online rewards of Section 6. TTTS tends to explore more than TS. Random and BC* primarily estimate coefficients, unlike the other policies which seek to optimize rewards.

8.2. Experimental Setup for the Two Applications

Sepsis is defined as a life-threatening organ dysfunction caused by dysregulated host response to infection and has been operationalized with an acute increase in the sequential (sepsis-related) organ failure assessment (SOFA) score of two points or more (Singer et al. 2016). Higher SOFA scores reflect greater organ dysfunction. For both applications, we use a configuration of model coefficients motivated by work on sepsis treatment to illustrate our proposed approach. We do not intend to provide medical recommendations. For more details about these applications and their modeling in our framework, see Online Appendix EC.7.1.

8.2.1. MARS Application: Sepsis Treatment Based on Transcriptomic Endotypes. Scicluna et al. (2017; MARS Consortium) use gene expression profiles from blood samples at admission and analyze them by unsupervised consensus clustering. They identify four endotypes (Mars1, Mars2, Mars3, and Mars4) for patients with sepsis and assess the association of these endotypes with clinical traits and survival outcomes. Later, van Mourik et al. (2022) study three aspects of treatment that may interact with one or more Mars endotypes in an exploratory study for personalizing treatment. Observational data suggest that some treatments may interact with some endotypes. Here, we explore the design of a potential trial to assess suspected interactions between endotypes and treatments.

We study the three aspects of treatment studied by van Mourik et al. (2022): fluid management (restrictive or liberal), tighter glucose control (no or yes), and hydrocortisone dose (low or high). This gives $n = 2^3 = 8$ treatment alternatives. We observe the Acute Physiology and Chronic Health Evaluation (APACHE) score at admission, a common measure of disease severity and widely used prognostic covariate for sepsis patients (Zimmerman et al. 2006). The primary outcome Y_t is taken to be the improvement (decrease) in SOFA score (Lambden et al. 2019) over a seven-day period following the diagnosis of sepsis. We approximate Y_t as a continuous quantity.

We choose a labeling that follows the exploratory findings of van Mourik et al. (2022). We assume that type Mars3 is predictive with respect to treatments with restrictive fluid management. A presumed treatment effect for one treatment with liberal fluid management ensures that some treatments are better than others for each Mars endotype. All endotypes are prognostic, consistent with Scicluna et al. (2017). We assume that the APACHE score at admission is prognostic and not predictive with respect to any treatment. Of 45 coefficients, $|\Xi| = 10$ are potentially active. Unless otherwise specified, for this example we test comparators on random problem instances. See Online Appendix EC.7.1.1 for more details.

Because we can efficiently compute the fEVI allocation policy when $\Delta = 0$ and the covariates are discrete and finite (using the method in Alban et al. (2021)), we first assume, for purposes of illustration, that $\Delta = 0$ (the patient interarrival time is larger than the delay in observing outcomes) and that the support of the APACHE score at admission is three possible values: low, medium, or high. We divide the APACHE score at the first and third quartile to obtain a discrete distribution that takes the value 0 (low) with probability 1/4, 1 (medium) with probability 1/2, and 2 (high) with probability 1/4. The probabilities for each of the types Mars1, Mars2, Mars3, and Mars4 are independent of the APACHE score at admission and are, respectively, proportional to 150, 184, 129, and 59; the counts for these types reported by Scicluna et al. (2017). Ensuing experiments explore trial performance with $\Delta > 0$.

8.2.2. Application on Lipkovich et al. Case Study: Sepsis Treatment with Clinical Data. Lipkovich et al. (2017, section 3.1) introduce a case study of a two-armed clinical trial ($n = 2$) for severe sepsis, which collected 11 covariates from clinical data (biomarkers such as age, APACHE II score, and baseline platelets). Using the outcomes of the trial retrospectively, Lipkovich et al. (2017, figures 8 and 9) present the results of a Lasso analysis that identifies which features are predictive, prognostic, or idle. The feature space included the treatment indicator, the 11 covariates, two-way biomarker interactions, interactions of the individual biomarkers with the treatment, and three-way interactions of pairs of covariates with the treatment. Of the 156 features in Lipkovich et al., 12 were found to be prognostic, 10 predictive (so that $|\Xi| = 22$), and the remaining idle. No label for treatment effect was identified. The values of the coefficients are provided in Online Appendix EC.7.1.2.

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Using these estimated coefficients to define a fixed instance setting, we test the performance of the policies with more covariates than the MARS application but only two treatments. We assume that the 11 original covariates are normally distributed. The trial's outcome was 28-day survival. For purposes of illustration in our experiment, we assume that the outcome is normally distributed with $\sigma = 0.5$ (the largest possible standard deviation for a Bernoulli random variable, given the outcome in the Lipkovich et al. trial was binary). To specify a prior distribution, we simulate a pilot study with 80 patients in each trial arm and use the robust prior described in Section 7.

8.3. *f*EVI Policy Has Strong Performance with Known Labeling for Offline Rewards

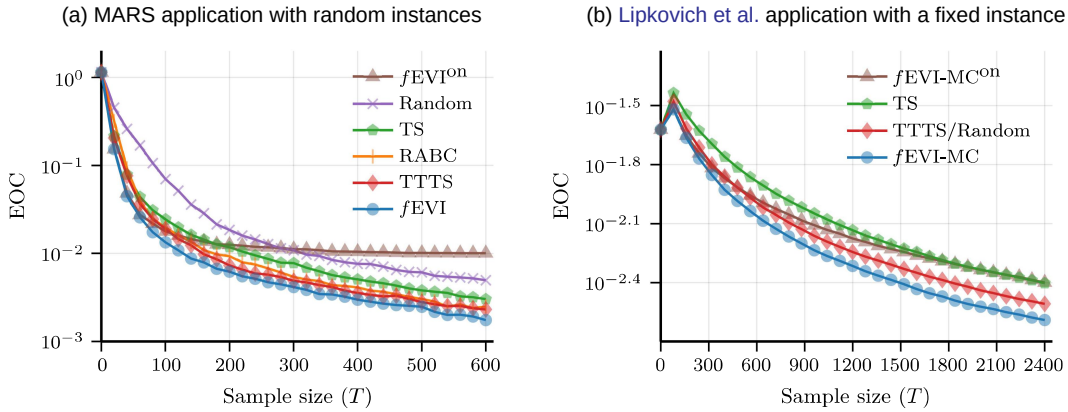
We compare the performance of the *f*EVI allocation policy, when the labeling is known, with that of comparator policies. Figure 2(a) shows the comparison for the MARS application. Curves show EOC as a function of the sample size for each policy. Lower curves are better. The *f*EVI policy obtains the lowest EOC. TTTS and RABC are closest but statistically worse. Here, $\text{EOC}^{f\text{EVI}}(600) = (1.75 \pm 0.14) \times 10^{-3}$, $\text{EOC}^{\text{TTTS}}(600) = (2.31 \pm 0.18) \times 10^{-3}$, $\text{EOC}^{\text{RABC}}(600) = (2.49 \pm 0.18) \times 10^{-3}$, where we report the average plus/minus one standard error of simulation results. TS is the next best ($\text{EOC}^{\text{TS}}(600) = (3.03 \pm 0.25) \times 10^{-3}$). We omit OCBA for clarity because of its very close performance compared with RABC in our experiments. BC* performs close to the Random allocation policy (slightly better for small sample sizes) and is omitted for clarity. Although the absolute reduction in EOC may appear small for $T = 600$, higher SOFA scores for sepsis are associated with higher mortality and poorer outcomes. Therefore, even a small reduction in EOC, and therefore SOFA, may result in some saved lives when considered in light of the annual incidence of sepsis. To obtain the same EOC as Random does with sample size 600 ($\text{EOC}^{\text{Random}}(600) = (4.97 \pm 0.30) \times 10^{-3}$), TS requires 392 observations, RABC 313, TTTS 299, and *f*EVI 245. Smaller sample sizes may result in smaller clinical trial costs.

Although BC* and Random both explore, their exploration is focused on estimating parameters rather than optimizing treatment strategies. This explains their poorer performance among policies that explore. OCBA explores while also optimizing, but its allocation scheme assumes that Mars3 is predictive with respect to every treatment. RABC suffers from the same shortcoming. TS and TTTS both balance exploration and exploitation, nevertheless performing better than other policies that seek to explore. Policy $f\text{EVI}^{\text{on}}$ is more focused on exploitation (it does not consider post-trial patients in its allocations) and has a lower rate of improvement than the others for larger sample sizes.

Figure 2(b) shows the EOC for several policies for the Lipkovich et al. application. Because the application considers two treatments, some policies are equivalent in performance (e.g., Random and TTTS with $\beta = 0.5$), or perform very close to TTTS (e.g., BC* and OCBA), and have been omitted in the figure. We find that, as with the *f*EVI policy in the MARS application, the *f*EVI-MC policy performs well in this setting, which, unlike the MARS application, is a fixed problem instance, has more covariates, and has fewer treatments. We observe that the sample size required to obtain the same EOC as that of the $f\text{EVI} - \text{MC}^{\text{on}}$ policy at $T = 2{,}400$ ($\text{EOC}(2{,}400) = (3.99 \pm 0.005) \times 10^{-3}$) is 2,377 for TS, 1,821 for TTTS/Random, and 1,484 for *f*EVI-MC. However, we note that the difference between *f*EVI-MC and Random is relatively smaller because this application only considers two treatments. One expects a bigger benefit with *f*EVI-MC in multi-arm trials. $f\text{EVI-MC}^{\text{on}}$ again has the worst EOC (it was designed for a different objective) but has a better rate of improvement compared with the MARS application.

Even though *f*EVI is designed to minimize EOC, it also performs well on another offline rewards

Figure 2. (Color online) EOC and Known Correct Labeling



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metric: the probability of correct selection (see Online Appendix EC.7.2).

8.4. MC Simulation Is Practically Useful for Estimating *f*EVI Indices

The usefulness of the proposed *f*EVI-MC algorithm, which uses MC simulation to estimate the *f*EVI indices, is threefold. First, *f*EVI-MC performs almost as well as *f*EVI, even with a small number of replications η^{on} and η^{off}. Figure 3(a) shows the EOC for the *f*EVI-MC allocation policy with several values of η^{off}. In this experiment, we fixed $\eta^{\text{on}} = 1$ because, when $\Delta = 0$, the *f*EVI-MC policy makes $\eta^{\text{on}}\eta^{\text{off}}$ i.i.d. simulations regardless of the specific values of η^{on} and η^{off}. The *f*EVI-MC policy performs very closely to *f*EVI even with a small number of sampled covariates and post-trial patients (small $\eta^{\text{off}}\eta^{\text{on}}$) for sample sizes considered in this figure.

Second, *f*EVI-MC avoids a numerical integration required by *f*EVI when there is positive delay, $\Delta > 0$, or when there is at least one continuous-valued patient covariate. Figure 3(b) shows EOC curves with delay $\Delta = 0, 20, 50$ for *f*EVI-MC for the MARS application, where we have replaced the APACHE score at admission with a normally distributed prognostic covariate with the same mean and standard deviation. The figure shows a modest, but not significant, degradation in EOC for *f*EVI-MC with delays over the tested range.[3] We also note that *f*EVI-MC can handle continuous covariates more effectively than using a version of *f*EVI that bins continuous covariates (Alban et al. 2024).

Third, because it can handle positive delay, *f*EVI-MC allows reaping the benefits of knowing the covariates and treatment allocations of pipeline patients before their outcomes are observed, when further treatment allocations are made. To see this, we introduce an allocation policy that uses data on patients whose outcomes have been observed, but not data on patients still in the pipeline. We call the latter allocation policy *f*EVI-MC-blind. Figure 3(b) shows two performance curves for *f*EVI-MC-blind with $\Delta = 20, 50$. Knowledge of the pipeline even before seeing the outcome, as done by *f*EVI-MC, is important and beneficial for arm allocation, particularly for up to sample sizes of four to five times the Δ. Ignoring the pipeline has a smaller negative effect if the sample size exceeds several multiples of Δ, after many samples have been observed.

We note that the *f*EVI-MC algorithm employs an important variance reduction, leveraged from the cKG (Frazier et al. 2009, Ding et al. 2021). This is accomplished in steps 11–13 of Algorithm 1 through the use of the $h(\cdot)$ function of cKG. That $h(\cdot)$ function enables exact computation of the integral over the next patient's outcome, conditional on simulated pipeline outcomes and covariates of post-trial patients.

Thus, we recommend *f*EVI-MC as a useful algorithm in contexts like ours. In experiments to date, *f*EVI-MC performs well even with values of $\eta^{\text{on}}, \eta^{\text{off}}$ as low as 5–10 each.

8.5. *f*EVI^{on} Policy for Online Rewards Has Strong Performance

Thus far, we have presented results evaluated through EOC and PICS, which are metrics of offline rewards: the rewards obtained from treating the post-trial population with the learned treatment strategy. We now present results related to online rewards: the rewards obtained from the patients in the trial. To measure the performance of online rewards, we report the cumulative regret, commonly used in the multi-armed bandit literature:

$$\text{cumulative regret}^{\pi}(\boldsymbol{\mu}, T) := \sum_{t=1}^{T} \mathbb{E}^{\pi}\left[\max_{w \in \mathscr{W}} (w \otimes \mathbf{X}_t)\boldsymbol{\mu} - (W_t \otimes \mathbf{X}_t)\boldsymbol{\mu} \,|\, \boldsymbol{\mu}\right]. \quad (19)$$

For a random instance setting, we average realizations of random problem instances, that is, cumulative $\text{regret}^{\pi}(T) = \mathbb{E}[\text{cumulative regret}^{\pi}(\boldsymbol{\mu}; T) \,|\, \boldsymbol{\theta}_0^{nat}, \Sigma_0^{nat}]$.

Figure 3. (Color online) EOC of Allocation Policies Using MC Estimates of the *f*EVI Indices

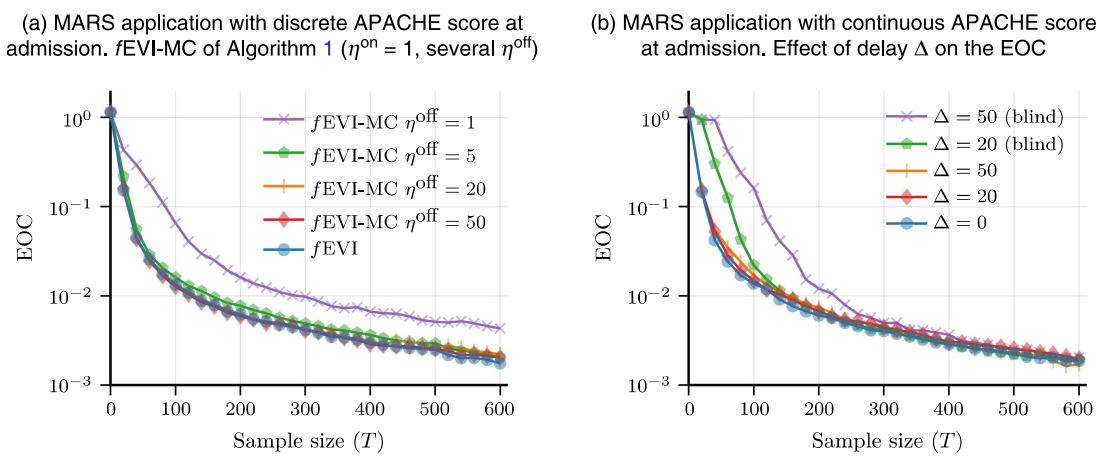

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Figure 4(a) shows the cumulative regret for the MARS application with the same allocation policies reported in Figure 2. Policies fEVI, RABC, and Random have a linear increase in cumulative regret because they are designed to explore. Figure 4(b) zooms into the remaining allocation policies. TTTS has a much lower, and apparently sublinear, cumulative regret compared with the allocation policies designed for exploration. Here, TTTS does reasonably well both for offline and online rewards. TS is designed to balance exploration and exploitation, is guaranteed to have a sublinear cumulative regret (Russo and Van Roy 2014), and performs better than TTTS in terms of this metric. $f\text{EVI}^{\text{on}}$ is designed to explore and exploit for the given sample size and obtains the lowest cumulative regret. Online Appendix EC.7.3 gives similar results for online rewards for the Lipkovich et al. application.

In the experiments of this section, we observed that fEVI has the best performance for offline rewards, and $f\text{EVI}^{\text{on}}$ has the best performance for online rewards, whereas their respective performance in the other metric is inferior to comparators. Moreover, TS and TTTS have strong performance in both offline and online rewards. In further experiments with the MARS application, we found that we can effectively trade off online and offline rewards using the $f\text{EVI}^{\text{on+off}}$ policy of Section 6. A weighted average of the online and offline fEVI indices, $f\text{EVI}^{\text{on+off}}$, attained comparable performance to that of TS and TTTS for both offline and online rewards. A decision maker could select the fEVI-based policy that best aligns with their objective. See Online Appendix EC.7.6 for numerical results.

8.6. Other Insights: Randomization and Performance on Challenging Fixed Instance

Randomization of patients to arms is desirable for clinical trial practice. However, many dynamic programming approaches, such as those we have used for fEVI, result in deterministic algorithms. We show that the fEVI-rand and fEVI-MC-rand allocation policies can introduce a moderate and controllable degree of randomness (e.g., randomizing up to 20% of patients) without significantly degrading EOC performance (Online Appendix EC.7.4). We also show that for a challenging fixed-instance, slippage-type configuration for the MARS application, the fEVI policy still does well compared with other policies (Online Appendix EC.7.5).

9. What Is the Cost of an Unknown Labeling? What Can Be Done?

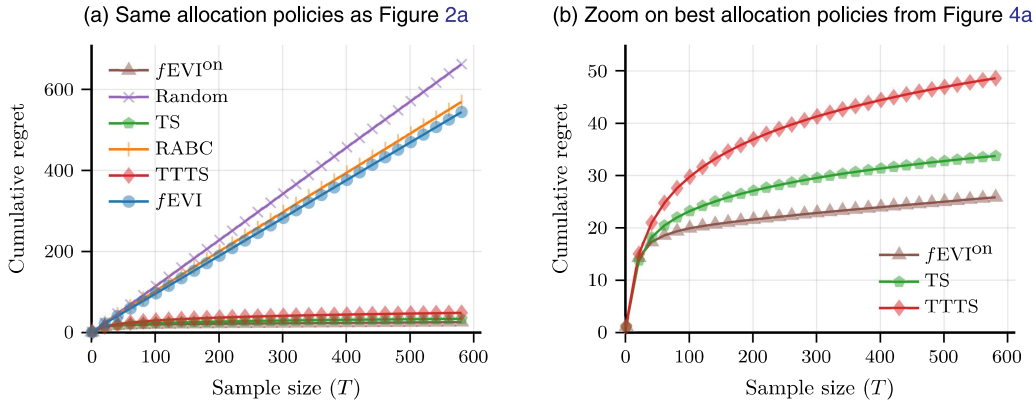
The experiments in Section 8 assume that the correct labeling of regression coefficients in (4) is known to the modeler. There are applications where such knowledge is not available. We discuss the potential costs of different ways of not knowing the correct labeling in Section 9.1. We then propose a heuristic to infer the labeling in Section 9.2.

9.1. Does Mislabeling Predictive and Prognostic Covariates Hurt Inference?

A lack of knowledge about the labeling of the covariates may impact the speed and quality of inference in a clinical trial. There are a combinatorial number of ways to mislabel covariates as being potentially predictive or prognostic: There are $2^{(n+1)(m+1)}$ choices for the $\xi_{i,l}$ in (4). Two interesting extreme mislabelings that may pose problems for any allocation policy are labelings where (i) all covariates are assumed to be predictive with respect to all treatments and not prognostic, or (ii) all covariates are assumed to be only prognostic. We present numerical results for these mislabelings with the case studies of Section 8.2.

To describe these extreme mislabelings, we let $\pi_{\text{all–prog}}$ denote an allocation policy π that assumes a labeling where all covariates may be prognostic but not predictive with respect to any treatment. We let $\pi_{\text{all–pred}}$ denote a policy π based on a labeling where all covariates are potentially predictive with respect to

Figure 4. (Color online) Cumulative Regret for the MARS Application with Known Labeling



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all treatments, but not prognostic (e.g., fEVI$_{\text{all–prog}}$ and fEVI$_{\text{all–pred}}$). For both mislabelings, if the known labeling contains no treatment effects then we include them so that a best treatment can be inferred. If the known labeling has treatment effects, we use those same treatment effects in the mislabelings, in order to have a direct comparison.

9.1.1. Unknown Personalization Potential (All-Prog Labeling). In the MARS example, suppose the trial manager is unaware that Mars3 is a predictive covariate and labels it as only prognostic; that is, all covariates are labeled only as prognostic. Figure 5, (a) and (b), shows that the EOC of fEVI$_{\text{all–prog}}$ approaches an asymptote (the best EOC that can be achieved without personalization), which is larger than the desired EOC of zero. Assuming all-prog implies that treatment effects for a population are averaged over subpopulations where the treatment works very well and not very well, and thus the value of precision medicine is not achieved.

9.1.2. What if All Covariates Are Potentially Predictive (All-Pred Labeling). Suppose the trial manager labels all covariates as being predictive with respect to all treatments, with no prognostic covariates. For the MARS example, this results in 42 instead of 10 coefficients to estimate. Figure 5(a) shows that this mislabeling slows down, but allows, inference. The EOC achieved by fEVI$_{\text{all–pred}}$ with sample size 600 is achieved by fEVI (with a known labeling) with only 90–100 observations: Knowing the correct labeling makes inference much faster. For the Lipkovich et al. example, which has many more features (156), the cost of assuming the all-pred labeling rather than the known labeling is also significant in Figure 5(b) for fEVI-MC. The fEVI$_{\text{all–pred}}$ allocation requires more than 1,000 observations before eventually improving on the EOC of the fEVI$_{\text{all–prog}}$ allocation because it has more parameters to estimate. Similar observations hold for the TS and Random policies (data not shown).

We explore through the MARS application the effect of mislabeling only a few terms. Our findings reinforce that mislabeling in the form of too many predictive terms has small losses in EOC compared with the losses when the mislabeling comes in the form of omitting any predictive term. Online Appendix EC.7.7 has further details.

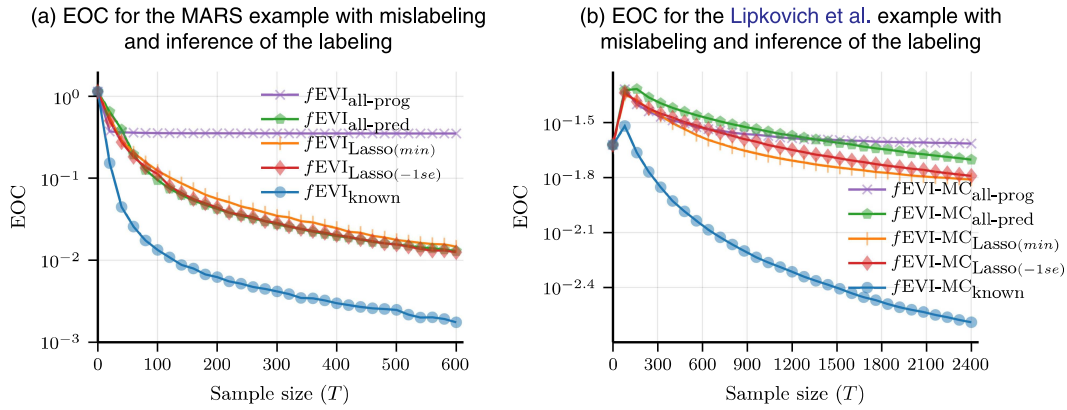
9.2. Can *f*EVI Be Enhanced to Allow the Labeling to Be Inferred?

Any of the allocation policies described in Sections 4, 6, and 8 assumes a specified labeling. We now introduce a heuristic to enhance any of these policies with model selection, in order to infer a useful labeling as patient outcomes are observed. We summarize the heuristic and experimental results here. Online Appendix EC.8 gives further details.

We assume that the modeler specifies an initial labeling with most (if not all) labels active in (4) and uses Lasso (Tibshirani 1996) to select a sparse set of active regressors (consistent with related work, e.g., Wang and Powell (2016) and Bastani and Bayati (2020)). As data are observed, we update the labeling from time to time: We found it useful to do so whenever t is a perfect square. Each allocation with a given policy is computed using the most recent updated labeling.

We use cross-validation to select the regularization parameter and allow for three variations: (1) *min* denotes the use of the regularization parameter that minimizes cross-validation error; (2) 1*se* denotes the use of the largest value of regularization parameter such that the cross-validation error is within one standard error of the minimum cross-validation error; and (3)−1*se* denotes the use of the smallest value of the regularization parameter such that the cross-validation error is within one standard error of the minimum cross-validation error.[4] We use $\pi_{\text{Lasso}(x)}$ (e.g., fEVI-MC$_{\text{Lasso}(x)}$) to denote the extension of a policy π that uses Lasso to

Figure 5. (Color online) Cost of Mislabeling Covariates and Potential Benefit of Model Selection to Regain Some of the Loss



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improve its best estimate of a sparse set of active regressors for a choice of the regularization parameter $x \in \{min, 1se, -1se\}$.

Figure 5(a) shows the performance of $f\text{EVI}_{\text{Lasso}(-1se)}$ and $f\text{EVI}_{\text{Lasso}(min)}$ for the MARS example: $f\text{EVI}_{\text{Lasso}(-1se)}$ performs statistically similar to the $f\text{EVI}_{\text{all-pred}}$ policy that assumes all covariates are predictive, despite the all-pred labeling being allowed an advantage over the Lasso(·) procedures by assuming a labeling with the correct treatment effects, and $f\text{EVI}_{\text{Lasso}(min)}$ performs slightly worse than $f\text{EVI}_{\text{all–pred}}$, and $f\text{EVI}_{\text{Lasso}(1se)}$ slightly worse than both of those (data not shown). Figure 5(b) shows the same analysis for the Lipkovich et al. case study. We see an improvement with the inferred labelings (Lasso($-1se$) and Lasso(min)) over the all-pred and all-prog labelings for the fEVI-MC policy. Overall, Lasso helps infer a sparse labeling from a large feature space and supports faster learning than with an all-pred labeling. However, knowing the true labeling gives a substantially lower EOC than does the inferred labeling.

Remark 3. The model selector for this Lasso proposal can also be used, at times chosen by a trial manager, as a diagnostic to help assess whether the assumed labeling is appropriate or whether the data suggest that an alternative model may be useful to consider.

10. Conclusion

The efficiency of clinical trials for precision medicine can be improved by leveraging insights about the labeling of patient covariates as predictive or prognostic. Such insights may come from observational data, as in the MARS case study, or from biomedical research that targets specific biomarkers. Our modeling and algorithmic contributions suggest methods to obtain these efficiencies for contexts where the goal is to best treat either post-trial patients (offline rewards) or in-trial patients (online rewards). Our numerical results suggest a promise for using our proposed fEVI family of allocation policies in clinical trial design for precision medicine, when insights about the predictive-prognostic covariate structure are available. We also identify the costs of not knowing a suitable covariate labeling and propose a practical method to address that concern. Online Appendix EC.9 discusses extensions of our model for further practical issues related to clinical trials. The framework here may be useful for optimal contextual learning applications beyond clinical trials.

As a practical recommendation, we recommend the fEVI-MC (or fEVI-MC-rand if more control of randomization is required) allocation policy for two reasons. First, it is quick to compute because it avoids numerical integration for evaluating allocation indices when outcomes are delayed or when some covariates have continuous support. Second, it is very effective due to an adaptation of results for the correlated knowledge gradient that gives a useful variance reduction for comparing indices.

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Endnotes

[1] Compare Lemma EC.3 in the Online Appendix to lemma A.7 of Frazier et al. (2009), lemma 7 of Xie et al. (2016), and proposition 3 of Ding et al. (2021).

[2] Recall Example 1. If all type A patients get treatment 1 and all type B patients get treatment 2, then both treatments are sampled infinitely often but one cannot learn the mean outcome for the two combinations never sampled (A-2, B-1).

[3] In contrast, delay significantly increases regret for fEVI-MC^{on} during the first Δ patient enrollments, leading to an upward shift in cumulative regret (results not shown).

[4] Choices min and $1se$ are standard and $-1se$ is a variation that we found to work well in our setting.

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